Comparative Verification and Synthesis of Human-Centric and Adaptive AI Systems

1. Systematic Claims Verification and Source Evaluation

Rigorous claim verification and source evaluation constitute the bedrock of evidence-based software engineering and Human-Computer Interaction (HCI) research. Establishing empirical truth through exact textual audits ensures that design guidelines, functional specifications, and architectural frameworks are grounded in verifiable findings rather than unvetted assumptions. In safety-critical contexts—such as emergency early warning systems or adaptive artificial intelligence (AI) decision-support tools—unverified assertions can lead to catastrophic system failures, compromised accessibility, and degraded user trust. The following evaluation matrices audit published research claims against primary source excerpts to establish empirical validity and identify analytical gaps.

1.1 NeuroAdaptX (Bunde) Claim Verification Matrix

The claims below were evaluated exclusively against the provided source text excerpt for *NeuroAdaptX: Designing Neuro-Adaptive Explanations for Cognitive Accessibility in Explainable AI Interfaces* (978-3-032-28316-0_12.pdf).

Claim Statement	**Verdict**	**Exact Source Wording**	**Page / Location Reference**
1. Adapts structure, modality, and density.	Confirmed	"...adapts structure, modality, and density to self-reported profiles while preserving informational equivalence."	Abstract / p. 206
2. Targeted at self-reported condition profiles (ADHD, autism spectrum conditions / ASC, dyslexia).	Confirmed	"...adapts presentation-level properties... based on self-reported neurodiversity profiles..."	Abstract / p. 206
3. Derives 10 design requirements from cognitive theory.	Partly	"...derive theory-grounded design requirements (DRs) and design principles (DPs)..."	Section 1 / p. 206
4. Evaluated via an online vignette experiment with N=216.	Partly	"...evaluated their effects against a static baseline in a randomized online experiment." (N = 216)	Abstract / p. 206
5. Utilizes a digital-wellbeing task context.	Not found	*N/A (No textual support in provided excerpt)*	N/A (Excerpt scope limitation)
6. Informational equivalence checked with	Not found	"...evaluating their effects... under informational equivalence."	Abstract / p. 206

Claim Statement	Verdict	Exact Source Wording	Page / Location Reference
a pre-deployment checklist.			

Analytical Note on Gaps: While Claim 3 confirms the theoretical derivation of design requirements (DRs), the exact numeric count of "10" is not specified in the excerpt. For Claim 4, the randomized online experiment with $N=216$ is confirmed, but the specific term "vignette" does not appear in the text provided. Claim 5 (digital-wellbeing context) and Claim 6 (pre-deployment checklist) lack explicit textual verification within the available context excerpt.

1.2 Tielman (2024) Claim Verification Matrix

The claims below were systematically audited against the provided source context repository.

Claim Statement	Verdict	Exact Source Wording	Page / Location Reference
1. Statement that "Adaptive XAI is the key to inclusive XAI".	Not found	*N/A (Source document absent from repository)*	N/A (Scope limitation)
2. Personalisation is "more than just simplicity in text".	Not found	*N/A (Source document absent from repository)*	N/A (Scope limitation)
3. People with cognitive disabilities should be involved throughout XAI development.	Not found	*N/A (Source document absent from repository)*	N/A (Scope limitation)
4. Counterfactuals are not necessarily feasible or actionable.	Not found	*N/A (Source document absent from repository)*	N/A (Scope limitation)
5. Most research presents feature importance.	Not found	*N/A (Source document absent from repository)*	N/A (Scope limitation)

Analytical Note on Gaps: Tielman (2024) is not present within the provided source context material. Under strict empirical source fidelity, an exact verdict of Not found is rendered across all evaluated claims due to context repository scope limitations.

1.3 Madugalla et al. Claim Verification Matrix

The claims below were audited against the primary source text for Madugalla et al., *Human-Centred Requirements Engineering for Critical Systems: Insights from Disaster Early Warning Applications* (2511.12856v2.pdf / arXiv version).

Claim Statement	Verdict	Exact Source Wording	Page / Location Reference
1. Addresses four vulnerable groups (older adults, low digital literacy, rural, colour-blind).	Confirmed	"...relevant to four vulnerable communities: older adults, low-digital-literacy users, rural users, and colour-blind users."	Abstract / p. 1
2. Translates 62 guidelines into 67 functional and non-functional requirements.	Confirmed	"...identified 62 guidelines... translated into a catalogue of 67 functional and non-functional requirements..."	Abstract / p. 1
3. Evaluated through 6 interviews and 8 evaluators across 4 personas.	Confirmed	"...evaluated through six interviews and eight cognitive walkthroughs."	Abstract / p. 1 & Section 8.2 / p. 12
4. Cognitive impairments are explicitly outside the scope.	Confirmed	"Other forms of vulnerability, such as linguistic diversity, cognitive impairments, and socioeconomic constraints, were outside the scope..."	Section 10.3 / p. 15
5. A fire danger forecast is shown but not explained.	Partly	"...Next four-day fire danger forecast for users' current area so that they could stay vigilant..."	Section 7.1 / p. 10
6. Adaptivity must remain explainable and controllable.	Confirmed	"...results also revealed that this adaptivity must always remain explainable and controllable..."	Abstract / p. 1 & Section 11 / p. 16

Analytical Audit & Source Contrast: The audited source context explicitly represents the arXiv version (2511.12856v2.pdf). A secondary Social Science Research Network (SSRN) version is absent from the provided source context. The textual analysis confirms the exact translation of 62 guidelines into 67 requirements across 4 target groups, validated via 6 community interviews and 8 walkthrough evaluators (2 per persona). Regarding Claim 5, while Section 7.1 confirms the inclusion of a four-day fire danger forecast feature (feature 2c), asserting that it is "not explained" represents an unverified extrapolation not supported by the primary text.The systematic verification of these claims establishes a solid factual foundation, enabling a seamless transition to synthesizing the broader research literature.

2. Synthesis of Related Work & Literature Integration

Synthesizing related work is vital for structuring complex academic domains, establishing software engineering baselines, and contextualizing novel empirical contributions. Rather than listing disconnected studies, structured synthesis highlights overarching methodological paradigms, identifies systemic design gaps, and clarifies open technical frontiers across human-computer interaction and requirements engineering.

2.1 NeuroAdaptX: Related Work Synthesis

NeuroAdaptX addresses cognitive heterogeneity in Explainable AI by adapting presentation-level properties—structure, modality, and density—for neurodiverse cognitive accessibility without altering substantive explanations. Grounded in Cognitive Load Theory and Cognitive Fit Theory within a Design Science Research framework, the approach formulates prescriptive design principles for adaptive interfaces. In a randomized between-subjects experiment ($N=216$), NeuroAdaptX significantly improved objective comprehension and trust while reducing perceived cognitive load compared to static baselines. However, the study relies on self-reported profiles during initial setup, leaving open research frontiers regarding real-time dynamic adaptation, expanding beyond self-reported profiles, and evaluating long-term performance across live, high-stress task environments.

2.2 Tielman (2024): Related Work Synthesis

Tielman (2024) frames adaptive Explainable AI (XAI) as a foundational paradigm for achieving meaningful cognitive inclusion across diverse user populations. Moving beyond rudimentary text simplification, the framework advocates for continuous participatory involvement of cognitively diverse stakeholders throughout all software development phases. The research highlights a critical disconnect in current literature, contrasting the prevailing research focus on feature importance metrics against the practical need for actionable, user-tailored explanation paradigms. Ultimately, the work leaves open fundamental questions regarding the real-world operational feasibility, actionability, and algorithmic safety of complex explanation methods—such as counterfactual explanations—when deployed for users with significant cognitive disabilities.

2.3 Madugalla et al.: Related Work Synthesis

Madugalla et al. establish a traceable human-centred requirements engineering process and validated requirements catalogue for inclusive disaster early warning systems. The multi-stage methodology translates 62 literature-derived guidelines into 67 functional and non-functional requirements, operationalised via an adaptive mobile prototype and evaluated through participatory feedback. Primary quantitative evaluations across four vulnerable groups—utilising 6 community interviews and 8 persona walkthrough evaluators—achieved 84% validation coverage and 70% positive validation. However, Figma prototype constraints left implementation-level non-functional requirements like offline performance and crash recovery unevaluated, while leaving excluded vulnerabilities, such as cognitive impairments and non-English linguistic diversity, for future research expansion.By synthesizing these contributions, a shared core theme emerges: adaptivity must balance automated user customization with direct human control, establishing the platform for a detailed cross-domain trade-off analysis.

3. Strategic Comparative Analysis and Cross-Domain Trade-Offs

Comparing adaptive Explainable AI (XAI) interfaces with human-centred requirements engineering (RE) for safety-critical systems reveals shared challenges in user agency, cognitive load, and visual layout. Both domains recognize that static, "one-size-fits-all" user interfaces fail vulnerable populations—whether neurodiverse individuals attempting to process complex AI explanations or emergency responders and citizens attempting to act on fire danger warnings under extreme stress. Cross-analyzing these domains highlights

essential architectural trade-offs between automated presentation management, explicit user control, and content preservation.

3.1 Architectural Trade-Offs: Adaptivity, Controllability, and Informational Equivalence

Evaluating presentation-level XAI adaptation against safety-critical early warning applications reveals fundamental differences in how adaptation is triggered, how content integrity is preserved, and how user mental models are maintained.

Architectural Dimension	**Presentation-Level XAI Adaptivity (NeuroAdaptX)**	**Safety-Critical Early Warning Systems (Madugalla et al.)**
Adaptivity Mechanisms	System-driven presentation adjustment modifying information structure, modality, and density based on self-reported profile settings.	User-driven self-customization allowing explicit selection of font sizes, navigation modes (scroll vs. page-based), language, and visual colour scales.
Preservation Constraints	**Informational Equivalence:** Mandates that underlying substantive explanation content remains identical; only visual structure and density change.	**Safety & Warning Clarity:** Mandates that critical warning content, danger ratings, and evacuation instructions remain uncompromised and instantly actionable across custom visual themes.
User Agency & Control	Profile-based adaptivity designed to reduce extraneous working memory load while maintaining calibrated trust and engagement.	Explicit user control over settings, ensuring adaptivity remains fully explainable and controllable to prevent disorientation during high-stress situations.

Theoretical Grounding: Cognitive Load, Cognitive Fit, and Disorientation

Applying Cognitive Load Theory (CLT) and Cognitive Fit Theory (CFT) explains why adaptivity mechanisms must differ fundamentally between decision support and emergency management contexts. In NeuroAdaptX, presentation-level adaptation reduces *extraneous working memory load* by matching information structure to a user's self-reported cognitive style (CFT), allowing intrinsic cognitive load to be dedicated to evaluating AI reasoning.However, in safety-critical early warning systems, unannounced or automated implicit interface reconfigurations pose severe operational risks. During an active emergency (e.g., an escalating bushfire alert), users experience extreme psychological stress and severe temporal pressure. If an interface dynamically alters its navigation layout, color encoding, or information density without explicit user initiation, it disrupts the user's established spatial mental model. This cognitive friction increases extraneous load at the precise moment when working memory capacity is severely constrained, inducing disorientation, decision paralysis, and potentially fatal delays.Consequently, as Madugalla et al. demonstrate, safety-critical adaptivity must remain strictly *user-driven, explainable, and controllable* . Providing explicit settings, in-context system notifications (e.g., confirming "Your colour scale has been updated"), and predictable linear navigation paths ensures that

interface adaptations preserve mental models, reinforce user agency, and maintain calibrated operational trust under stress.

3.2 Methodological Frameworks for Vulnerability Elicitation and Validation

Evaluating software accessibility across vulnerable communities requires structured methodologies that balance direct qualitative stakeholder feedback with systematic analytical walkthroughs.

Methodological Trade-Offs: Stakeholder Interviews vs. Persona Walkthroughs

The empirical evaluation framework in Madugalla et al. highlights key methodological trade-offs between qualitative field research and analytical proxy evaluations:

1. **Direct Community Stakeholder Interviews ($N=6$):** Conducted with older adults ($n=2$) and rural residents ($n=4$), direct interviews capture rich, contextual lived experiences. For example, interviews revealed critical real-world usability friction, such as external links to Google Maps causing navigation entrapment for older adults who struggled to return to the core application. However, direct engagement suffers from severe recruitment bottlenecks, geographic constraints, and high resource overhead when attempting to reach isolated populations.
2. **Persona-Based Cognitive Walkthroughs ($N=8$ across 4 personas):** Utilizing expert evaluators paired with detailed user personas enables systematic, scalable interface testing across constrained target groups (e.g., colour-blind or low-digital-literacy users). While highly effective at identifying structural usability defects, persona walkthroughs are inherently synthetic; they cannot fully simulate real-time cognitive degradation, physical hand tremors, motor skill decline, or network dropouts during an active crisis.

Requirements Breakdown and Intersectional "Curb-Cut" Analysis

Madugalla et al. translated 62 literature guidelines into 67 functional and non-functional requirements (NFRs). Analyzing the category distribution reveals a heavy prioritization of quality constraints:

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                REQUIREMENT DISTRIBUTION BY CATEGORY

Usability NFRs       [============================] 27
Accessibility NFRs   [===================] 19
Functional Reqs      [=================] 17
Performance NFRs     [===] 3
Reliability NFRs     [=] 1
```

A critical insight emerges when examining the **intersectional overlap** across user groups. Out of 67 requirements, **23 (34.3%) are shared across multiple target populations** :

- **Elderly + Rural (9 shared requirements, R48–R56):** Focuses on extended interaction durations, clear visual focal points, visible return paths, and text alternatives.
- **Elderly + Rural + Low-Digital-Literacy (6 shared requirements, R62–R67):** Focuses on flat navigation hierarchies (≤ 2 levels deep), location clarity, single consistent task routes, and concise error messages.

- **Rural + Low-Digital-Literacy (5 shared requirements, R57–R61):** Focuses on lightweight media optimization, shared-device support, low data usage, and offline caching.
- **Elderly + Colour-Blind (3 shared requirements, R45–R47):** Focuses on popup contrast, grayscale luminance testing, and WCAG AA/AAA compliance.This 34.3% intersectional overlap provides empirical evidence of the **"curb-cut effect" in Requirements Engineering** . Designing explicitly for intersecting vulnerabilities reduces redundant system specifications while systematically elevating universal usability for all end users.

Empirical UI Friction Findings

During prototype testing (Section 8.3), qualitative feedback identified specific usability friction points that illustrate the necessity of human-centred RE:

- *Map Symbol Ambiguity:* Participants routinely mistook the blue dot representing their current location for a nearby regional town named "Orange" due to lack of a map legend.
- *Topographical Misinterpretation:* Dotted danger boundary lines were misinterpreted as topographical contour lines due to missing visual keys.
- *Anxiety Over Emergency Controls:* Participants expressed severe anxiety regarding unconfirmed speed-dial buttons for emergency contacts (000 / SES), requesting modal confirmation dialogues to prevent accidental dials during high-stress handling.

Requirement Traceability Matrix Analysis & Figma Prototype Limitations

The overall evaluation achieved an **84% validation coverage rate** (56 of 67 requirements addressed) and a **70% positive validation rate** (47 of 67 confirmed).

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                    VALIDATION OUTCOMES (N=67)

 Positively Validated [========================>    ] 70% (47)
 Addressed (Total)    [============================> ] 84% (56)
 Not Evaluated        [=====>                       ] 16% (11)
```

However, **11 requirements (16%) remained unevaluated** . A technical audit of Table 10 and Section 8.3.5 explains why medium-fidelity Figma prototypes are inherently incapable of evaluating these specific non-functional requirements:

1. **Performance NFRs (R57, R59, R60):** Interactive UI prototypes cannot simulate network packet delivery, server-side image compression algorithms, or low-bandwidth data throttling under degraded 3G/4G rural coverage.
2. **Connectivity & Offline NFRs (R34, R58, R61):** Figma canvas environments cannot test OS-level background SMS fallback protocols, local storage caching routines, or peer-to-peer package distribution when app stores are unreachable.
3. **Reliability NFRs (R29):** Static prototype simulations cannot simulate operating system application crashes, memory leaks, or state recovery routines.
4. **Hardware & Accessibility NFRs (R20, R56):** Prototypes cannot evaluate mobile OS screen-reader alt-text parsing efficiency or hardware-level acoustic emergency alerts.These unverified properties demonstrate that while interactive visual prototypes are optimal for eliciting layout and navigation feedback, definitive technical assurance for safety-critical systems requires fully deployed software builds tested on

physical hardware in live, low-connectivity field environments.In conclusion, embedding human-centricity into requirements engineering and adaptive AI system architecture ensures that accessibility, user control, and technical reliability operate as integrated operational dimensions of safety and resilience.